

Student's Name: _____ Teacher's Code: _____



Saint Ignatius' College, Riverview
Mathematics Assessment Task
2024

Year 12
Mathematics (Extension One)
Task 4
Trial HSC Examination
Date: 21 st August 2024

General Instructions:

- **Reading time:** 10 minutes
- **Time Allowed:** 2 hours.
- Write using a black pen.
- Calculators approved by NESA may be used.
- Attempt all questions in the booklets provided.
- Write **your name** and **your teacher's code** in the positions indicated.
- Marks may not be awarded for missing or carelessly arranged work.

Teachers:

- | | |
|----------------|------------|
| • Mr R Maxwell | REM |
| • Mr D Reidy | DPR |
| • Mr N Mushan | NHM |
| • Mr J Newey | JPN |

Topics Examined:

Section A **10 Marks**
Multiple Choice

Section B

Short Answer	
Question 11	15 Marks
Question 12	15 Marks
Question 13	15 Marks
Question 14	15 Marks
Total	70 Marks

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SECTION I: Multiple Choice Questions:

1. How many integer solutions are there for the inequality $\frac{1}{|x-2|} > \frac{1}{4}$?

- A. 7
- B. 6
- C. 5
- D. 4

2. Consider the equation $3x^3 + 2x^2 - x + 5 = 0$.

What is the sum of the reciprocals of the roots of this equation?

- A. $\frac{1}{5}$
- B. $\frac{2}{5}$
- C. $\frac{5}{2}$
- D. -5

3. What is the maximum value of $15\sin\theta - 8\cos\theta + 2$?

- A. 9
- B. 17
- C. 19
- D. 23

4. Find the cartesian equation of the curve represented by the parametric equations:

$$x = 4 - t \text{ and } y = 3t^3.$$

- A. $y = 64 - 48x + 12x^2 - x^3$
B. $y = 64 + 48x + 12x^2 + x^3$
C. $y = 192 + 144x + 36x^2 - 3x^3$
D. $y = 192 - 144x + 36x^2 - 3x^3$

5. A particle is initially at $x = -1$. Its motion has a velocity of $v = \frac{1}{t + e}$.

Which of the following will be true when the particle is at the origin?

- A. $t = e^{-1}, \quad v = 1$
B. $t = 1, \quad v = e^{-1}$
C. $t = e^{-2}, \quad v = -1$
D. $t = e^2 - e, \quad v = e^{-2}$

6. What is the domain and range of $y = 4\cos^{-1} \frac{3x}{2}$?

- A. Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$
Range: $-4\pi \leq y \leq 4\pi$.
B. Domain: $-\frac{3}{2} \leq x \leq \frac{3}{2}$
Range: $-4\pi \leq y \leq 4\pi$.
C. Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$
Range: $0 \leq y \leq 4\pi$.
D. Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$
Range: $0 \leq y \leq 4$.

7. The angle between two vectors $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ is approximately.

- A. $\cos^{-1}(.5)$
- B. $\cos^{-1}(-.5)$
- C. $\cos^{-1}(1)$
- D. $\cos^{-1}(-1)$

8. X, Y , and Z are collinear points, with position vectors $\underline{\mathbf{x}}$, $\underline{\mathbf{y}}$ and $\underline{\mathbf{z}}$ respectively.
 Y lies between X and Z .

Given that $|\overrightarrow{YZ}| = \frac{1}{2} |\overrightarrow{XY}|$, which of the following expressions is equal to $\underline{\mathbf{z}}$?

- A. $\frac{1}{2} \underline{\mathbf{x}} - \frac{3}{2} \underline{\mathbf{y}}$
- B. $\frac{3}{2} \underline{\mathbf{x}} - \frac{1}{2} \underline{\mathbf{y}}$
- C. $\frac{3}{2} \underline{\mathbf{y}} - \frac{3}{2} \underline{\mathbf{x}}$
- D. $\frac{3}{2} \underline{\mathbf{y}} - \frac{1}{2} \underline{\mathbf{x}}$

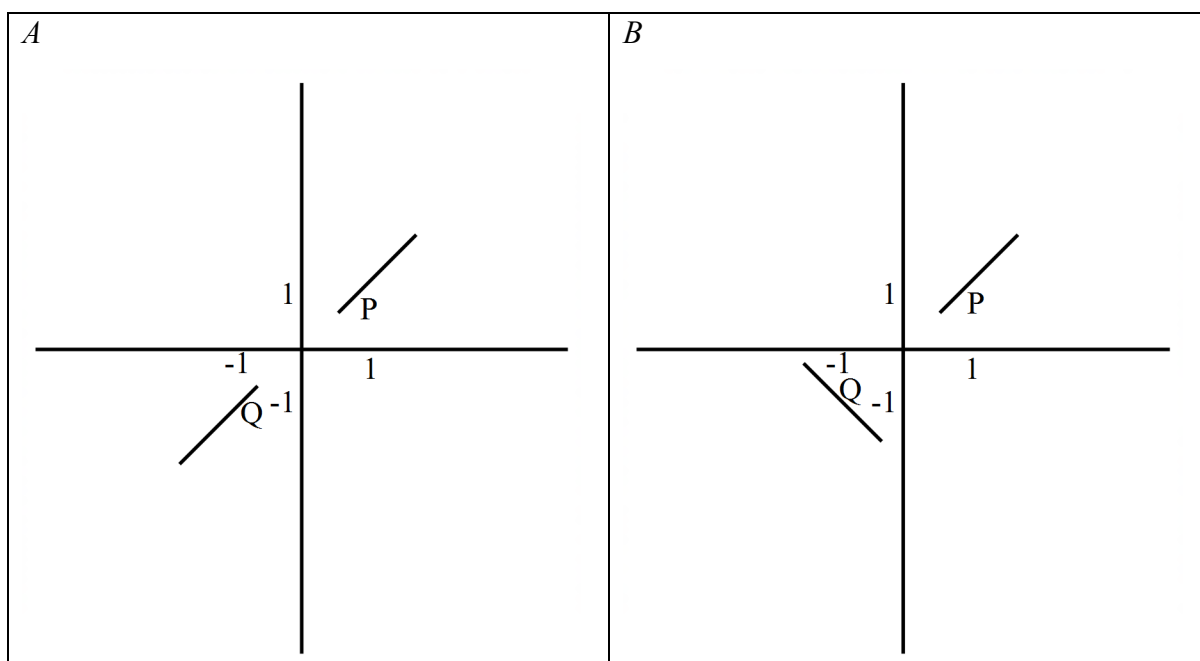
9. Which of the following is the derivative of $y = x^2 \sin^{-1}(2x)$?

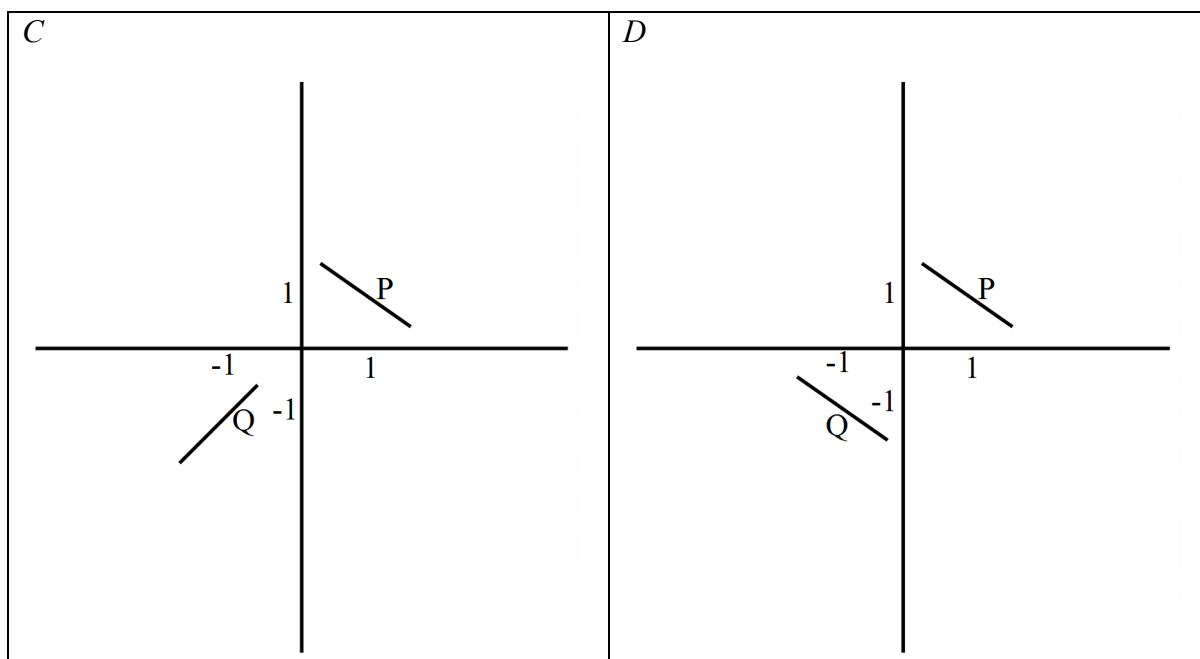
- A. $2x \sin^{-1}(2x) + \frac{2x^2}{\sqrt{1-4x^2}}$
- B. $2x \sin^{-1}(2x) + \frac{x^2}{\sqrt{1-4x^2}}$
- C. $2x \sin^{-1}(2x) + \frac{2x^2}{\sqrt{1-2x^2}}$
- D. $2x \sin^{-1}(2x) + \frac{x^2}{\sqrt{1-2x^2}}$

10. A directional field is to be drawn for the differential equation,

$$\frac{dy}{dx} = 2x + \frac{y}{x}.$$

Which of the following graphs shows the correct slope lines at the points $P(1,1)$ and $Q(-1,-1)$?





End of Section 1

Section II

60 marks

Attempt Questions 11-14

Answer each question in the appropriate writing booklet.

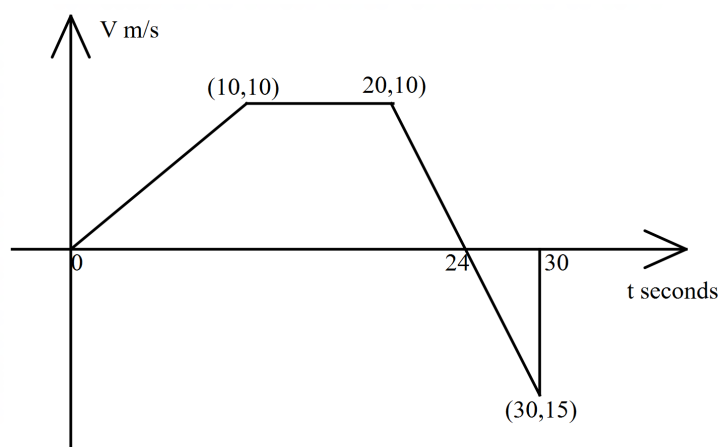
Extra exam writing booklets are available.

Question 11

Start on a new answer booklet

15 Marks.

- (a) Solve $|1 - 2x| \geq 15$. 2
- (b) Given the velocity time graph below find:
- (i) The acceleration for the first 10 seconds 1
 - (ii) The total distance travelled in the first 30 seconds. 1



- (c) Use the substitution $u = e^{2x}$ to find $\int \frac{e^{2x}}{9 + e^{4x}} dx$. 3
- (d) Water is poured into a container at a rate of $8 \text{ cm}^3/\text{s}$. If the volume of the water in the container is given by $V = \frac{3}{2}(h^2 + 8h) \text{ cm}^3$ where h is the depth of the water, find the rate of change of the depth of the water when $V = 72 \text{ cm}^3$. 4
- (e) (i) Express $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$ where $R > 0$ and $0 \leq x \leq 2\pi$. 2
- (ii) Find the co-ordinates of the points of intersection of the graph $y = \cos x - \sqrt{3} \sin x$ with the x and y axis, in the interval $0 \leq x \leq 2\pi$. 2

End of Question 11

Question 12

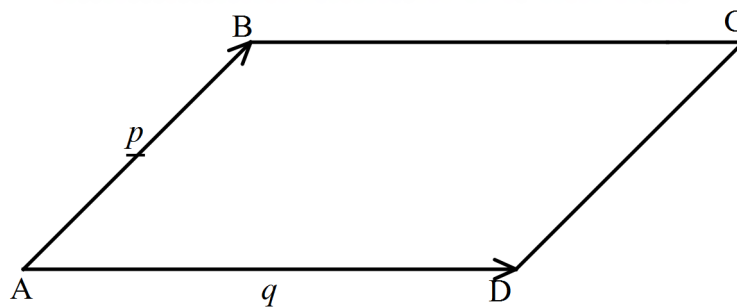
Start on a new answer booklet

15 Marks.

- (a) Prove that $\frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x} = 2$. 3
- (b) Use the substitution $x = 2\sin \theta$ to find the exact value of $\int_0^1 \sqrt{4 - x^2} dx$. 3
- (c) The vectors $\underline{\mathbf{u}} = \begin{pmatrix} 3 \\ \sin 2\alpha \end{pmatrix}$ and $\underline{\mathbf{v}} = \begin{pmatrix} \cos \alpha \\ -3 \end{pmatrix}$, where $0 < \alpha < \pi$, are perpendicular. Find possible values of α . 3

- (d) Let $P(x) = x^5 - 6x^3 - 8x^2 - 3x$.
Show that $x = -1$ is a root of $P(x)$ of multiplicity three. 3

- (e) $ABCD$ is a parallelogram, where $\overrightarrow{AB} = \underline{\underline{\mathbf{p}}}$ and $\overrightarrow{AD} = \underline{\underline{\mathbf{q}}}$.



Prove, using vectors, that the sum of the squares of the lengths of the diagonals is equal to the sum of the squares of the lengths of all the four sides. 3

End of Question 12

Question 13 Start on a new answer booklet 15 Marks.

- (a) (i) Show that $\sin(x - y)\sin(x + y) = \sin^2 x - \sin^2 y$. 1
- (ii) Hence or otherwise,
Solve $\sin^2 3x - \sin^2 x = \sin 4x$. $x = [0, \pi]$ 2
- (b) (i) Sketch the graph of $y = f(x) = x^2 - 1$. 1
- (ii) On the same graph sketch $\frac{1}{x^2 - 1} + 2$.
Clearly identifying asymptotes and intercepts. 3
- (c) Given $|\underline{\underline{\mathbf{p}}}| = 3$, $|\underline{\underline{\mathbf{q}}}| = 5$ and $\underline{\underline{\mathbf{p}}} \cdot \underline{\underline{\mathbf{q}}} = 6$.
Calculate the length of $2\underline{\underline{\mathbf{p}}} - 3\underline{\underline{\mathbf{q}}}$. 3

(d) Let $f(x) = \ln\left(\frac{2x}{3}\right)$.

(i) Show that the equation of the normal to the graph of $y = f(x)$ at the point

where $x = \frac{3e^2}{2}$ has the equation $y = -\frac{3e^2x}{2} + \frac{9e^4}{4} + 2$.

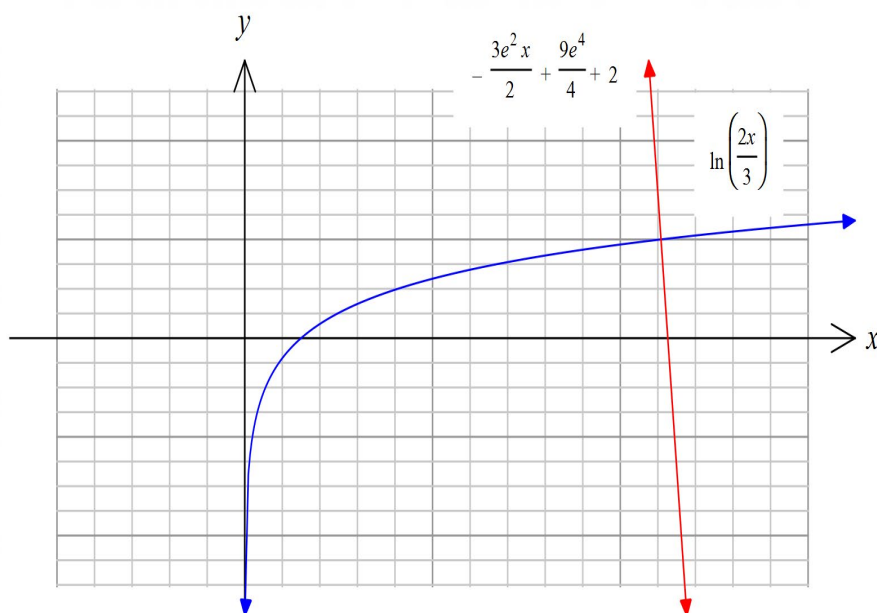
2

(ii) Find the exact area enclosed by the graph of $y = f(x)$, the line

$y = -\frac{3e^2x}{2} + \frac{9e^4}{4} + 2$ and the x axis.

3

(You may use the diagram below to help solve the problem)



End of Question 13

Question 14

Start on a new answer booklet

15 Marks.

(a) (i) Show that $\frac{d}{dx}(\tan^3 x) = 3\sec^4 x - 3\sec^2 x$.

1

(ii) Hence or otherwise, evaluate $\int_0^{\pi/4} \sec^4 x dx$.

3

(b) Prove by mathematical induction for all integers $n \geq 1$

that $\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n}$.

3

- (c) Given that $\frac{dy}{dx} = \frac{1}{4}(y - 1)^2$ and $y(0) = 0$.
What is the value of x when $y = 2$? 3
- (d) Solve the differential equation $\frac{dy}{dx} = \frac{2}{x^3 e^y}$ given $y(1) = 0$.
Express your solution in the form $y = f(x)$. 3
- (e) Let $\vec{g} = |\vec{e}| \vec{f} + |\vec{f}| \vec{e}$ where $\vec{e}$, $\vec{f}$ and $\vec{g}$ are non-zero vectors.
Show that $\vec{g}$ bisects the angle between $\vec{e}$ and $\vec{f}$. 2

End of Question 14

END OF TASK

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2024 Trial HSC Examination
Mathematics Extension 1 Course

Name Solutind'

Teacher REM DPR NHM JPN

Section I – Multiple Choice Answer Sheet

Allow about 15 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider the correct answer, indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ ^{correct}
C ☐ D ☐

1. A ☒ B ☐ C ☐ D ☐
2. A ☒ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☒ D ☐
4. A ☐ B ☐ C ☐ D ☒
5. A ☐ B ☐ C ☐ D ☒
6. A ☐ B ☐ C ☒ D ☐
7. A ☒ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☒
9. A ☒ B ☐ C ☐ D ☐
10. A ☐ B ☒ C ☐ D ☐

Start your answer here.

Question 11

(a) Solve $|1-2x| \geq 15$

$$1-2x = -15$$

$$1-2x = 15$$

$$-2x = -16$$

$$-2x = 14$$

$$x = 8$$

$$x = -7$$



$$\therefore x \leq -7$$

U

$$x \geq 8$$

Many did not understand how to manipulate the inequalities

ie $1-2x \geq 15$ or $-1+2x \geq 15$

1 mark for each correct solution

(b) (i) $a = \frac{v}{t} = \frac{10}{10} = 1 \text{ m/s}^2$

1 mark

(ii) Distance travelled is Area A+B

$$A = \text{trapezium} = \frac{1}{2} (10+24) \times 10$$

$$= \frac{1}{2} 34 \times 10 = 170$$

$$B: \text{Triangle} = \frac{1}{2} \times 6 \times 15 = 45$$

Distance not displacement

$\therefore$ total area above and below axis

$\therefore$ Distance travelled is $170+45$

$$= 215 \text{ m.}$$

1 mark

$$(c) u = e^{2x} \quad du = 2e^{2x} dx$$

$$\frac{1}{2} du = e^{2x} dx$$

$$\therefore \int \frac{e^{2x}}{9+e^{2x}} dx = \frac{1}{2} \int \frac{du}{9+u^2}$$

$$= \frac{1}{2} \int \frac{du}{3^2+u^2} = \frac{1}{6} \int \frac{3 du}{3^2+u^2}$$

$$= \frac{1}{6} \tan^{-1}\left(\frac{u}{3}\right) + C$$

$$= \frac{1}{6} \tan^{-1}\left(\frac{e^{2x}}{3}\right) + C.$$

correct organisation

correct integral in terms of u

some did not realise that it involved inverse tan

correct answer

(d) Given $\frac{dv}{dt} = 8 \text{ cm}^3/\text{s}$ $V = \frac{3}{2}(h^2+8h) \text{ cm}^3$
Find $\frac{dh}{dt}$, when $V = 72 \text{ cm}^3$

$$\frac{dv}{dt} = \frac{dv}{dh} \times \frac{dh}{dt} \quad \frac{dv}{dh} = \frac{3}{2}(2h+8)$$

$$= 3h+12$$

Finding correct expression for $\frac{dv}{dh}$

$$\therefore 8 = (3h+12) \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{8}{3(h+4)}$$

correct Expression for $\frac{dh}{dt}$

To find 'h' $V = \frac{3}{2}(h^2+8h) = 72$

$$h^2+8h = 48 \quad h^2+8h-48 = 0$$

$$(h-4)(h+12) = 0 \quad h = 4, h = -12 \therefore h = 4$$

finding value of h

$$\therefore \frac{dh}{dt} = \frac{8}{3(4+8)} = \boxed{\frac{1}{3} \text{ cm/s.}}$$

calculating value of $\frac{dh}{dt}$

Additional writing space on back page.

ex) Express $\cos x - \sqrt{3} \sin x \equiv R \cos(x+\alpha)$

$$1 \cos x - \sqrt{3} \sin x$$

$$R = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

1 finding r

$$\tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3} \quad (\text{acute angle only})$$

$$\therefore \cos x - \sqrt{3} \sin x = 2 \cos\left(x + \frac{\pi}{3}\right)$$

1 finding α

(ii)

For points of intersection:

$$\text{at } x=0 \quad y = 2 \cos\left(\frac{\pi}{3}\right) = 2 \times \frac{1}{2} = 1$$

$$(0, 1)$$

finding y

$$\text{at } y=0 \quad 2 \cos\left(x + \frac{\pi}{3}\right) = 0$$

intercept

$$\cos\left(x + \frac{\pi}{3}\right) = 0$$

$$0 \leq x \leq 2\pi$$

$$x + \frac{\pi}{3} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \dots$$

$$x = \frac{\pi}{2} - \frac{\pi}{3}, \frac{3\pi}{2} - \frac{\pi}{3}, \frac{5\pi}{2} - \frac{\pi}{3}, \frac{7\pi}{2} - \frac{\pi}{3}, \dots$$

$$x = \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}, \frac{19\pi}{6}, \dots$$

out of domain

$$\therefore x = \frac{\pi}{6}, \frac{7\pi}{6}$$

finding co-ords

of x intercepts

$$\therefore \left(\frac{\pi}{6}, 0\right), \left(\frac{7\pi}{6}, 0\right)$$

Start your answer here.

Question 12

$$(a) \frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x}$$

$$\sin x \cos x$$

$$= \frac{1 \times 2 \sin x \cos x}{2}$$

$$= \frac{\sin 3x \cos x - \cos 3x \sin x}{\sin x \cos x} = \frac{\sin(3x-x)}{\frac{1}{2} \sin 2x}$$

$$= \frac{1}{2} \sin 2x$$

$$= \frac{2 \sin 2x}{\sin 2x} = 2$$

* Well Answered

* Most students split

$\sin 3x$ and $\cos 3x$ into

$\sin(2x+x)$ and $\cos(2x+x)$

and use appropriate

Trig identities

$$(b) x = 2 \sin \theta$$

$$dx = 2 \cos \theta d\theta \quad \text{when } x=0, 2 \sin \theta = 0, \theta = 0$$

$$x=1, 2 \sin \theta = 1 \quad \sin \theta = \frac{1}{2}, \theta = \frac{\pi}{6}$$

$$\therefore \int_0^1 \sqrt{4-x^2} dx = \int_0^{\pi/6} \sqrt{4-4\sin^2 \theta} \times 2 \cos \theta d\theta$$

* Poorly answered.

$$= \int_0^{\pi/6} 2 \cos \theta \times 2 \cos \theta d\theta$$

* Many students did not express dx in

$$= 4 \int_0^{\pi/6} \cos^2 \theta d\theta = 4 \times \frac{1}{2} \int_0^{\pi/6} 1 + \cos 2\theta d\theta$$

terms of $d\theta$

* Many students did

$$\therefore 2 \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/6}$$

$$= 2 \left[\frac{\pi}{6} + \frac{\sqrt{3}}{2 \times 2} - 0 - 0 \right]$$

not change the limits.

$$= 2 \left[\frac{\pi}{6} + \frac{\sqrt{3}}{4} \right] = \frac{\pi}{3} + \frac{\sqrt{3}}{2}$$

(c) $u \cdot v = 0$ for perpendicular

$$\begin{pmatrix} 3 \\ \sin 2\alpha \end{pmatrix} \cdot \begin{pmatrix} \cos \alpha \\ -3 \end{pmatrix} = 0$$

$$3 \cos \alpha - 3 \sin 2\alpha = 0$$

$$3 \cos \alpha - 6 \sin \alpha \cos \alpha = 0$$

$$3 \cos \alpha (1 - 2 \sin \alpha) = 0$$

$$\cos \alpha = 0, \quad 1 - 2 \sin \alpha = 0$$

$$\cos \alpha = 0 \quad \sin \alpha = \frac{1}{2}$$

$$\text{For } \cos \alpha = 0 \quad \alpha = 0, \pi/2$$

$$\text{For } \sin \alpha = \frac{1}{2} \quad \alpha = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\therefore \alpha = \pi/6, \pi/2, 5\pi/6$$

* Well answered.

* Some students

included $\frac{3\pi}{2}$ in

their solutions and forgot the limit for α

was $0 < \alpha < \pi$

* Many students forget

to include $\frac{5\pi}{6}$ in

the solution set.

(d) $P(x) = x^5 - 6x^3 - 8x^2 - 3x$

$$P'(x) = 5x^4 - 18x^2 - 16x - 3$$

$$P''(x) = 20x^3 - 36x - 16$$

To prove multiplicity of roots.

$$P(x) = P'(x) = P''(x) \text{ at } x = -1 \text{ (Given).}$$

$$\therefore P(-1) = -1 + 6 - 8 + 3 = 0$$

$$P'(-1) = 5 - 18 + 16 - 3 = 0$$

$$P''(-1) = -20 + 36 - 16 = 0$$

$\therefore x = -1$ is a root of multiplicity three.

or $(x+1)^3$ is a factor of $P(x)$.

* Nearly well answered.

* Some students

divided $P(x)$ by $x+1$

and then divided

the divided by $x+1$

and repeated the process until such

time as they were

able to express in

the form $(x+1)^3 Q(x)$

If you did this

correctly you get

full marks.

Additional writing space on back page.

$$(e) \quad \overline{AD} = \overline{BC} = \underline{q}$$

$$\overline{AB} = \overline{DC} = \underline{p}$$

$$\overline{AC} = \underline{p+q}, \quad \overline{BD} = \underline{p-q}$$

$\therefore$ Sum of Squares of Diagonal

$$|\underline{p+q}|^2 + |\underline{p-q}|^2$$

$$= (\underline{p+q}) \cdot (\underline{p+q}) + (\underline{p-q}) (\underline{p-q})$$

$$= \underline{p^2} + \underline{2pq} + \underline{q^2} + \underline{p^2} - \underline{2pq} + \underline{q^2}$$

$$= \underline{p^2} + \underline{q^2} + \underline{p^2} + \underline{q^2}$$

= Sum of the squares of all four sides.

* Well answered.

Start your answer here.

Question 13

(a) (i) $\sin(x-y) \sin(x+y)$

$$= \frac{1}{2} [\cos[(x-y)-(x+y)] - \cos[(x-y)+(x+y)]]$$

$$= \frac{1}{2} [\cos(-2y) - \cos(2x)]$$

$$= \frac{1}{2} (\cos 2y - \cos 2x)$$

$$= \frac{1}{2} [1 - 2\sin^2 y - (1 - 2\sin^2 x)]$$

$$= \frac{1}{2} [-2\sin^2 y + 2\sin^2 x]$$

$$= \sin^2 x - \sin^2 y.$$

(ii) From (i) $\sin^2 3x - \sin^2 x$

$$= \sin(3x-x) \sin(3x+x)$$

$$= \sin 2x \sin 4x$$

$$\therefore \sin^2 3x - \sin^2 x = \sin 4x$$

$$\text{ie } \sin 2x \sin 4x = \sin 4x \text{ from (i)}$$

$$\sin 2x \sin 4x - \sin 4x = 0$$

$$\sin 4x (\sin 2x - 1) = 0$$

$$\text{ie } \sin 4x = 0 \quad \sin 2x = 1$$

$$\text{For } \sin 4x = 0 \quad 4x = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$$

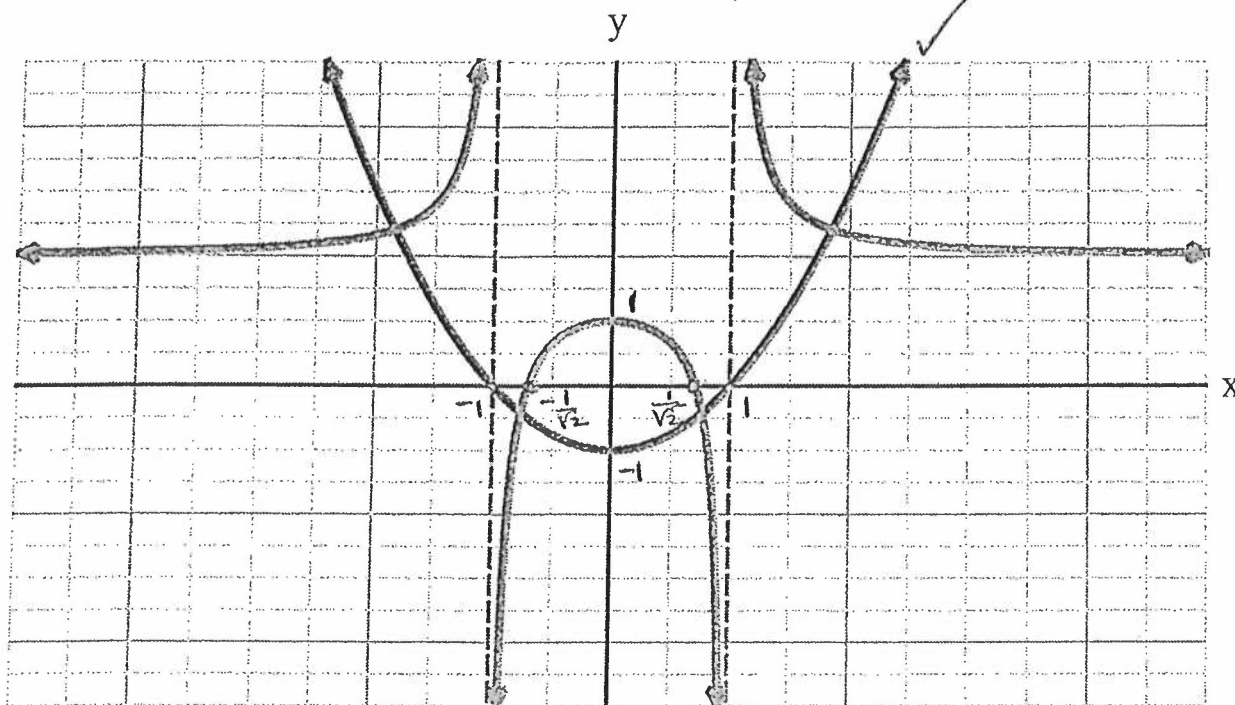
$$\sin 2x = 1, \quad 2x = \frac{\pi}{2}, \quad x = \frac{\pi}{4}$$

$$\therefore \text{Solution: } x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi.$$

Solutions to 13 (b) (i) and (ii)

(i) Sketch the graph of $y = f(x) = x^2 - 1$.

(ii) On the same graph sketch $\frac{1}{x^2 - 1} + 2$.



for x -intercepts.

$$\frac{1}{x^2 - 1} + 2 = 0$$

$$\frac{1}{x^2 - 1} = -2$$

$$1 = -2(x^2 - 1)$$

$$1 = -2x^2 + 2$$

$$\therefore 2x^2 = 1$$

$$x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}}$$

NOTE:

Most students
lost a mark for
not getting the
 x intercept

1 Mark Correct Shape

1 " Asymptotes

$$x = \pm 1$$

$$y = 2$$

1 Mark correct x
intercepts

$$(c) |p| = 3, |q| = 5, p \cdot q = 6$$

$$\text{Formula } |a - b| = \sqrt{(a-b)(a-b)}$$

$$\therefore |2p - 3q| = \sqrt{(2p - 3q) \cdot (2p - 3q)}$$

$$= \sqrt{2p \cdot 2p - 2p \cdot 3q - 3q \cdot 2p + 3q \cdot 3q}$$

$$= \sqrt{4(p \cdot p) - 6(p \cdot q) - 6(q \cdot p) + 9(q \cdot q)}$$

$$p \cdot p = |p|^2 = 3^2 = 9$$

$$q \cdot q = |q|^2 = 5^2 = 25$$

$$p \cdot q = 6$$

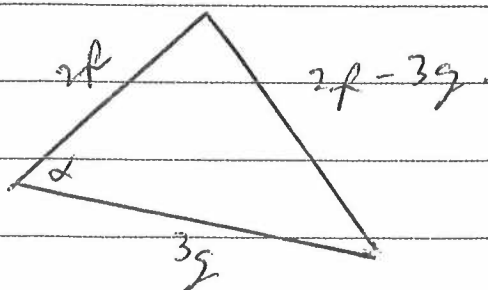
$$= \sqrt{4(9) - 12(6) + 9(25)}$$

$$= \sqrt{189}$$

$$\therefore |2p - 3q| = \sqrt{189} = \sqrt{9 \times 21}$$

$$= 3\sqrt{21}$$

(OR)



$$|2p - 3q|^2 = |2p|^2 + |3q|^2 - 2|2p||3q|\cos\alpha$$

$$= 4|p|^2 + 9|q|^2 - 2 \times 2|p| \times 3|q| \cos\alpha$$

$$= 4(3)^2 + 9(5)^2 - 12|p||q|\cos\alpha$$

$$= 189$$

$$- 2 \times 2 \times (3) \times 3 \times (5) \cdot \frac{6 \times 6}{2 \times 3 \times 3 \times 5}$$

$$\text{length } 2p - 3q = \sqrt{189}$$

$$= 3\sqrt{21}$$

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(d) $f(x) = \ln\left(\frac{2x}{3}\right)$

(i) $y = \ln 2x - \ln 3$

$$y' = \frac{1}{2x} \times 2 = \frac{1}{x}$$

$$m_t = \frac{1}{x} = \frac{2}{3e^2}$$

$$\therefore m_n = -\frac{3e^2}{2}$$

$$y = \ln\left(\frac{2x}{3}\right)$$

$$\text{at } x = \frac{3e^2}{2}$$

$$y = \ln\left(\frac{2 \times \frac{3e^2}{2}}{3}\right) = \ln e^2$$

$$= 2 \ln e = 2$$

1 Mark

Eqn. of normal. at $\left(\frac{3e^2}{2}, 2\right)$ is:

$$y - y_1 = m_n(x - x_1)$$

$$y - 2 = -\frac{3e^2}{2}(x - \frac{3e^2}{2})$$

$$y = -\frac{3e^2}{2}x + \frac{9e^4}{4} + 2$$

1 Mark

(ii) Area of shaded region

See diagram.

$$= A(\text{Trapezium OABC}) - \int_0^2 \frac{3}{2}e^y dy$$

$$y = -\frac{3e^2}{2}x + \frac{9e^4}{4} + 2$$

$$0 = -\frac{3e^2}{2}x + \frac{9e^4}{4} + 2$$

$$\frac{3e^2}{2}x = \frac{9e^4}{4} + 2$$

$$x = \frac{2}{3e^2} \times \frac{9e^4}{4} + \frac{2 \times 2}{3e^2}$$

$$x = \frac{3e^2}{2} + \frac{4}{3e^2}$$

$$A(\text{Trap}) = \frac{1}{2} \left(\frac{3e^2}{2} + \frac{3e^2}{2} + \frac{4}{3e^2} \right) \times 2$$

$$= 3e^2 + \frac{4}{3e^2}$$

Additional writing space on back page.

$$y = \ln\left(\frac{3x}{3}\right)$$

$$e^y = \frac{3x}{3}$$

$$3e^y = x$$

$$\text{A curve} = \int_0^2 \frac{3e^y}{2} dy$$

$$= \left[\frac{3e^y}{2} \right]_0^2 = \frac{3e^2}{2} - \frac{3}{2} \checkmark$$

$$1. \text{ Area Shaded} = \frac{3e^2 + 4}{3e^2} - \left(\frac{3e^2}{2} - \frac{3}{2} \right) \checkmark$$

$$= \frac{3e^2 + 4}{3e^2} - \frac{3e^2}{2} + \frac{3}{2}$$

$$= \frac{3e^2}{2} + \frac{4}{3e^2} + \frac{3}{2} u^2$$

$$y = \ln\left(\frac{2x}{3}\right)$$

$$y' = \frac{1 \times 2 \times 2}{2x \times 3} = \frac{1}{x}$$

$$m_t = \frac{1}{x} = \frac{2}{3e^2}$$

$$m_n = -\frac{3e^2}{2}$$

$$y = \ln\left(\frac{2x}{3}\right)$$

$$y = \ln\left(\frac{2}{3} \times \frac{3e^2}{2}\right)$$

$$y = \ln e^2$$

$$y = 2$$

$$y - 2 = -\frac{3e^2}{2} \left(x - \frac{3e^2}{2}\right)$$

$$y = -\frac{3e^2 x}{2} + \frac{9e^4}{4} + 2$$

✓
✓

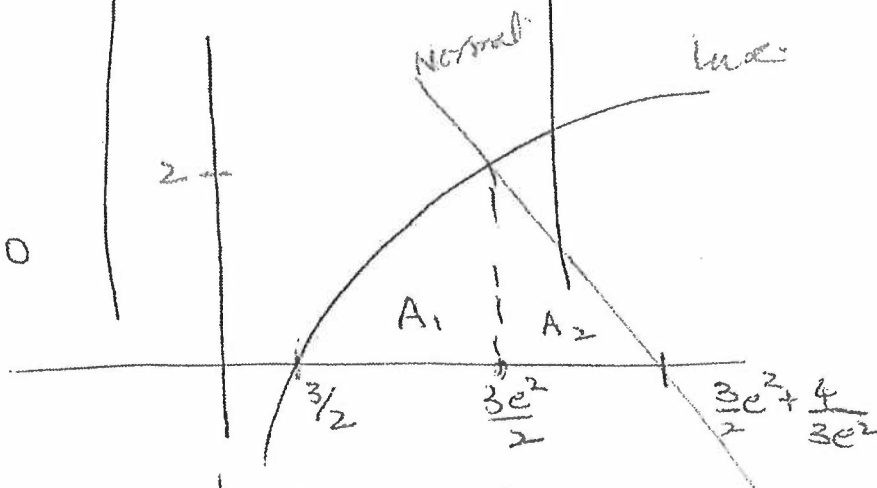
Area

at $y=0$ Curve.

$$\ln\left(\frac{2x}{3}\right) = 0$$

$$e^0 = \frac{2x}{3}$$

$$x = \frac{3}{2}$$



at $y=0$ Normal.

$$\frac{3e^2 x}{2} = \frac{9e^4}{4} + 2$$

$$x = \frac{\frac{3 \times 2 \times e^4}{3 \times \frac{1}{2}} + \frac{2 \times 2}{3e^2}}{e^2}$$

$$x = \frac{3e^2}{2} + \frac{4}{3e^2}$$

at $y=2$ Curve.

$$2 = \ln\left(\frac{2x}{3}\right)$$

$$e^2 = \frac{2x}{3}$$

$$x = \frac{3e^2}{2}$$

$$\text{Area} = A_1$$

$$\int_{3/2}^{3e^2/2} \ln\left(\frac{2x}{3}\right) dx + \frac{1}{2} \times b \times h.$$

$$+ \frac{1}{2} \left(\frac{3e^2}{2} + \frac{4}{3e^2} - \frac{3e^2}{2} \right) \times 2$$

$$+ \frac{4}{3e^2}$$

Int. by Parts.

Next Page.

$$A_1: \int_{3/2}^{3e^2/2} \ln\left(\frac{2x}{3}\right) dx$$

$$= \int \ln 2x - \ln 3 dx$$

$$= \int (\ln x + \ln 2 - \ln 3) dx$$

$$\therefore \left[x \ln x - x + x \ln 2 - x \ln 3 \right]_{3/2}^{3e^2/2}$$

$$\frac{3e^2}{2} \ln\left(\frac{3e^2}{2}\right) - \frac{3e^2}{2} + \frac{3e^2}{2} \ln 2 - \frac{3e^2}{2} \ln 3$$

$$- \frac{3}{2} \ln\left(\frac{3}{2}\right) + \frac{3}{2} - \frac{3}{2} \ln 2 + \frac{3}{2} \ln 3$$

$$= \frac{3e^2}{2} (\ln 3 + 2 - \ln 2) - \frac{3e^2}{2} + \frac{3e^2}{2} \ln 2 - \frac{3e^2}{2} \ln 3$$

$$- \frac{3}{2} (\ln 3 - \ln 2) + \frac{3}{2} - \frac{3}{2} \ln 2 + \frac{3}{2} \ln 3$$

$$= \frac{3e^2}{2} \ln 3 + \frac{3e^2}{2} - \frac{3e^2}{2} \ln 2 - \frac{3e^2}{2} + \frac{3e^2}{2} \ln 2 - \frac{3e^2}{2} \ln 3$$

$$- \frac{3}{2} \ln 3 + \frac{3}{2} \ln 2 + \frac{3}{2} - \frac{3}{2} \ln 2 + \frac{3}{2} \ln 3$$

$$= \frac{3e^2}{2} + \frac{3}{2}$$

$$= A_1 + A_2$$

$$= \frac{3e^2}{2} + \frac{3}{2} + \frac{4}{3e^2}$$

$$\int \ln x dx$$

$$= x \ln x - x$$

$$\int \ln 2 = x \ln 2$$

$$\int \ln 3 = x \ln 3$$

$$\ln\left(\frac{3e^2}{2}\right)$$

$$= \ln 3e^2 - \ln 2$$

$$= \ln 3 + \ln e^2 - \ln 2$$

$$= \ln 3 + 2 - \ln 2$$

QUESTION 14

$$\begin{aligned} \text{a) i) } \frac{d}{dx} (\tan^3 x) &= 3 \tan^2 x (\sec^2 x) \\ &= 3(\sec^2 x - 1)(\sec^2 x) \\ &= 3\sec^4 x - 3\sec^2 x \end{aligned}$$

Surprisingly, many
mocked this up.

Imk

$$\text{(ii) } \int_0^{\frac{\pi}{4}} \sec^2 \theta = ??$$

From (i)

$$3\sec^4 x - 3\sec^2 x = \frac{d}{dx} (\tan^3 x)$$

$$3\sec^4 x = \frac{d}{dx} (\tan^3 x) + 3\sec^2 x \leftarrow \text{Imk}$$

$$\sec^4 x = \frac{1}{3} \left[\frac{d}{dx} (\tan^3 x) \right] + \sec^2 x$$

$$\int_0^{\frac{\pi}{4}} \sec^4 x dx = \left[\frac{1}{3} \tan^3 x + \tan x \right]_0^{\frac{\pi}{4}} \leftarrow \text{Imk}$$

$$= \left(\frac{1}{3} \tan^3 \frac{\pi}{4} + \tan \frac{\pi}{4} \right) - (0)$$

$$= \frac{1}{3} + 1$$

$$= \frac{4}{3}$$

$\leftarrow$ Imk

Poorly done

Many students
"lost" the 3.

b) Prove $\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n}$
for $n > 1$

Step 1. Prove true for $n=1$.

$$\begin{aligned} \text{LHS} &= \frac{1}{2} \\ \text{RHS} &= 2 - \frac{3}{2} \\ &= \frac{1}{2} \end{aligned} \quad \therefore \text{true for } n=1$$

Imk for correct
Setting out

Step 2. Assume true for $n=k$.

$$\text{i.e. } \frac{1}{2} + \frac{2}{2^2} + \dots + \frac{k}{2^k} = 2 - \frac{k+2}{2^k}$$

Now Prove true for $n=k+1$

$$\text{i.e. } \frac{1}{2} + \frac{2}{2^2} + \dots + \frac{k+1}{2^{k+1}} = 2 - \frac{k+3}{2^{k+1}}$$

Imk for correct
Setting out

Proof.

$$\text{LHS} = \frac{1}{2} + \dots + \frac{k}{2^k} + \frac{k+1}{2^{k+1}}$$

$$= 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}} \quad \text{From assumption}$$

$$= \frac{2(2^{k+1}) - 2(k+2) + k+1}{2^{k+1}}$$

$$= \frac{2(2^{k+1})}{2^{k+1}} - \frac{k+3}{2^{k+1}}$$

$$= 2 - \frac{k+3}{2^{k+1}}$$

etc by
m.I

Imk for use
of assumption
& correct
working to
soln.

$$e) \frac{dy}{dx} = \frac{1}{4} (y-1)^2$$

METHOD 1

$$\frac{dy}{(y-1)^2} = \frac{1}{4} dx$$

$$(y-1)^{-2} dy = \frac{1}{4} dx$$

$$\frac{(y-1)^{-1}}{-1} = \frac{x}{4} + C$$

$$-\frac{1}{y-1} = \frac{x}{4} + C$$

$$x=0 \quad y=0$$

$$1 = C$$

$$-\frac{1}{y-1} = \frac{x}{4} + 1$$

$$y=2 \quad x=???$$

$$-\frac{1}{1} = \frac{x}{4} + 1$$

$$x = -8$$

METHOD 2

$$\frac{dx}{dy} = 4(y-1)^{-2}$$

$$x = -4(y-1)^{-1} + C$$

$$x = -\frac{4}{y-1} + C$$

$$x=0 \quad y=0$$

$$0 = -\frac{4}{-1} + C$$

$$C = -4$$

$$x = -\frac{4}{y-1} - 4$$

$$y=2 \quad x=??$$

$$x = -\frac{4}{1} - 4$$

$$= -8$$

link for either
correct method

link for correct
constant.

(many forgot
to allow for
constant)

link for
correct
solution

$$d) \frac{dy}{dx} = \frac{2}{x^3 e^y} \quad x=1 \quad y=0$$

$$e^y dy = \frac{2}{x^3} dx$$

$$e^y dy = 2x^{-3} dx$$

$$e^y = \frac{2x^{-2}}{-2} + C$$

$$e^y = -\frac{1}{x^2} + C$$

$$x=1 \quad y=0$$

$$e^0 = -1 + C$$

$$C = 2$$

$$\therefore e^y = -\frac{1}{x^2} + 2$$

$$y = \ln \left(2 - \frac{1}{x^2} \right)$$

link for correct method.

Again, many forgot the "C"

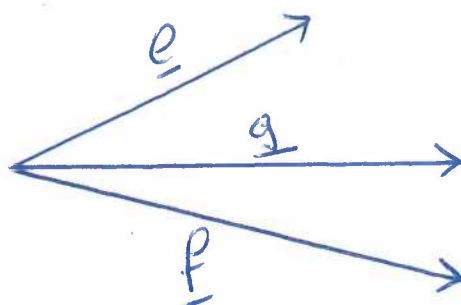
link for value of "C".

link for correct solution

e) One path to a correct solution is using the dot product

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$$

$$\underline{g} = |\underline{e}| \underline{f} + |\underline{f}| \underline{e}$$



$$\begin{aligned} \underline{e} \cdot \underline{g} &= \underline{e} \cdot [|\underline{e}| \underline{f} + |\underline{f}| \underline{e}] \\ &= |\underline{e}| \underline{e} \cdot \underline{f} + |\underline{f}| \underline{e} \cdot \underline{e} \\ &= |\underline{e}| \underline{e} \cdot \underline{f} + |\underline{f}| |\underline{e}|^2 \\ &= |\underline{e}| [\underline{e} \cdot \underline{f} + |\underline{f}| |\underline{e}|] \end{aligned}$$

Mark awarded
for substantial
effort

$$\therefore |\underline{e}| \cdot |\underline{g}| \cos \alpha = |\underline{e}| [\underline{e} \cdot \underline{f} + |\underline{f}| |\underline{e}|]$$

$$\text{OR } |\underline{g}| \cos \alpha = \underline{e} \cdot \underline{f} + |\underline{f}| |\underline{e}| \quad *$$

$$\text{Similarly, } \underline{f} \cdot \underline{g} = \underline{f} [|\underline{e}| \underline{f} + |\underline{f}| \underline{e}]$$

$$= |\underline{e}| |\underline{f}|^2 + |\underline{f}| \underline{e} \cdot \underline{f}$$

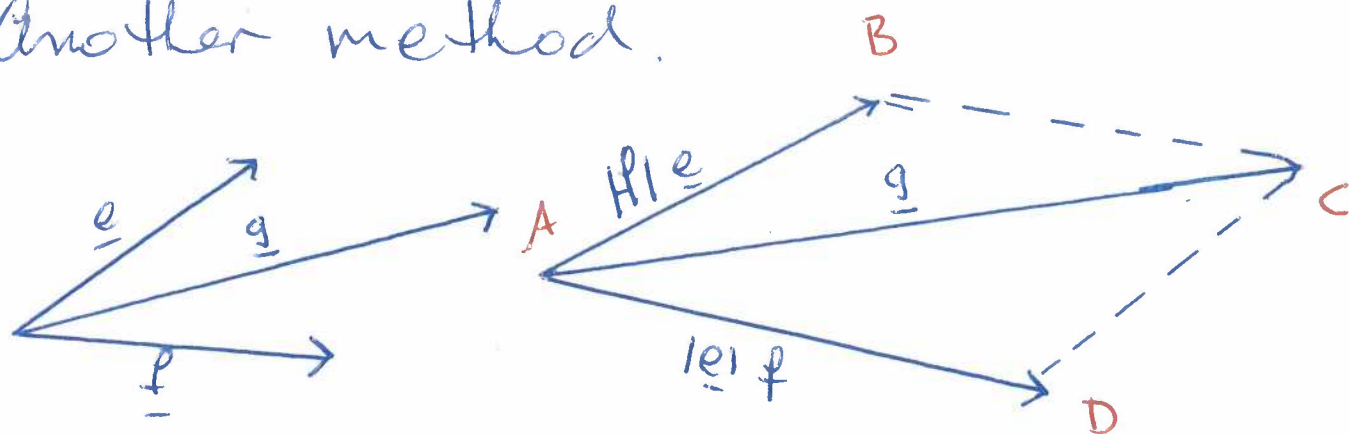
$$= |\underline{f}| [|\underline{e}| |\underline{f}| + \underline{e} \cdot \underline{f}]$$

$$\therefore |\underline{f}| |\underline{g}| \cos \beta = |\underline{f}| [|\underline{e}| |\underline{f}| + \underline{e} \cdot \underline{f}]$$

$$|\underline{g}| \cos \beta = |\underline{e}| |\underline{f}| + \underline{e} \cdot \underline{f} \quad *$$

$$\therefore \cos \alpha = \cos \beta$$

e) Another method.



$$\underline{g} = |\underline{f}| \underline{e} + |\underline{e}| \underline{f}$$

$$\text{Now } |\underline{f}| \underline{e} = |\underline{e}| \underline{f}$$

* MUST BE
REFERENCED
IN SOLUTION

$\therefore$ $ABCD$ is a Rhombus

If $ABCD$ is a Rhombus

then $\underline{g}$ bisects the angle between

$\underline{e}$ and $\underline{f}$.

Hence, it bisects the angle between

$\underline{e}$ and $\underline{f}$

1mk awarded
for substantial
effort